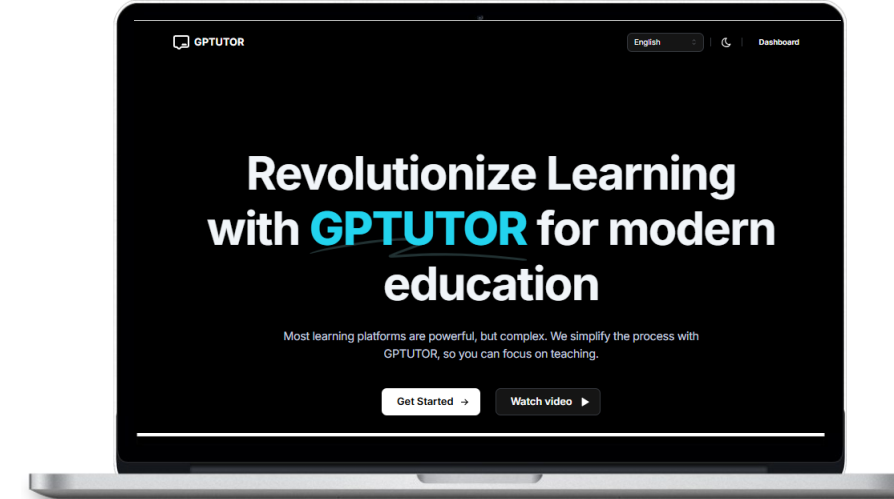


GPTutor Project

GPTutor

- A GenAI-powered Intelligent tutoring system (ITS) platform developed by our team at HK PolyU
- Features
 - Create courses, manage students, import learning scenarios
 - Upload learning resources (e.g. lecture slides)
 - Summarize contents and generate review exercises
 - Conversation interface for Q/A for the learning resources
 - Students may provide feedback to the AI's response
 - Instructor Dashboard
 - Learning analytics
 - Review student conversation history and interaction with the ITS
 - Provide customized answer to students' questions
 - Simulation and Role-Playing using GenAI

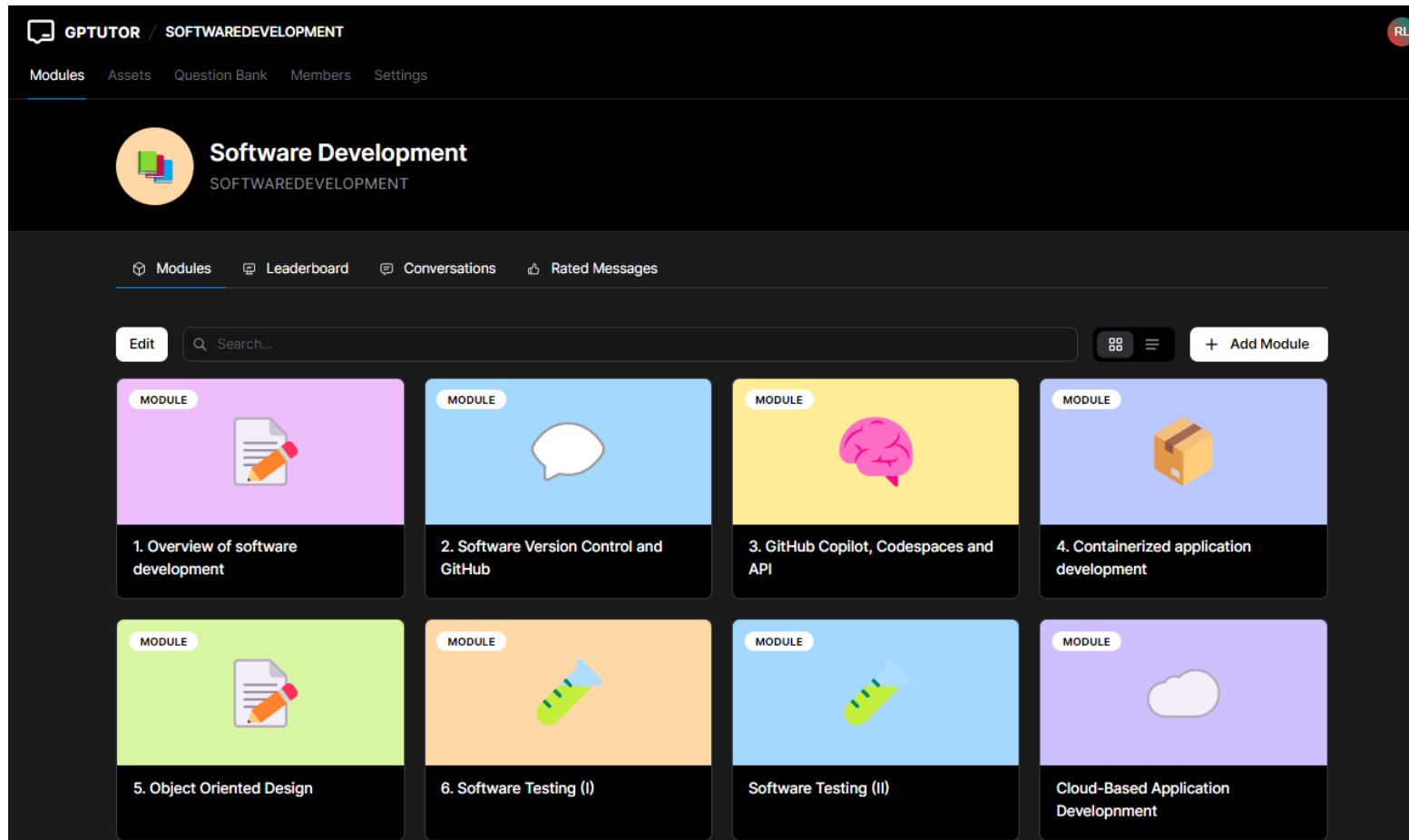


Our website:

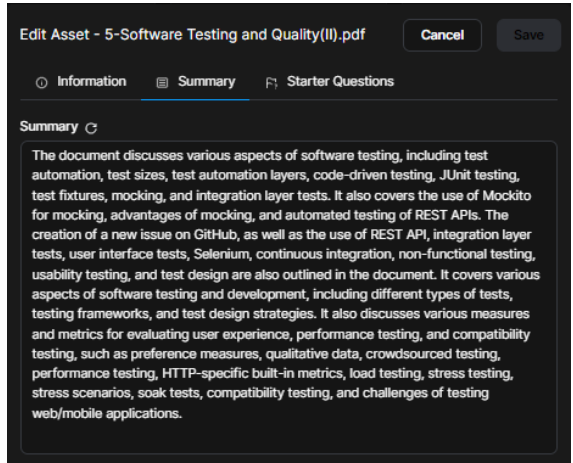
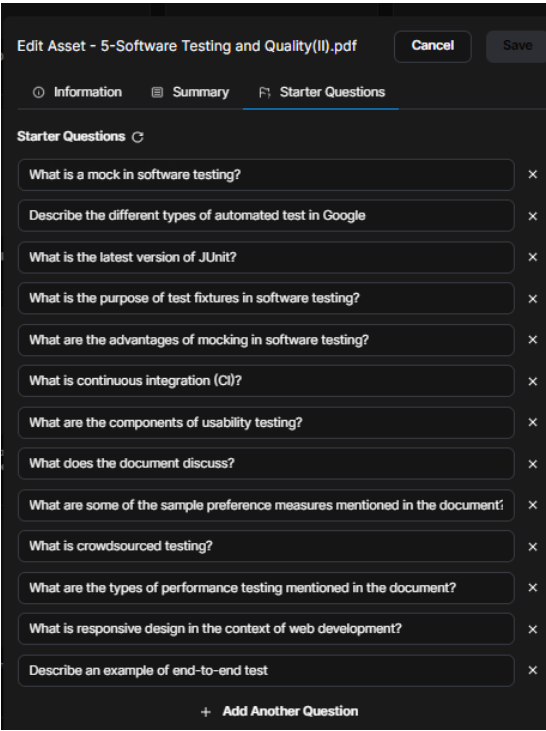
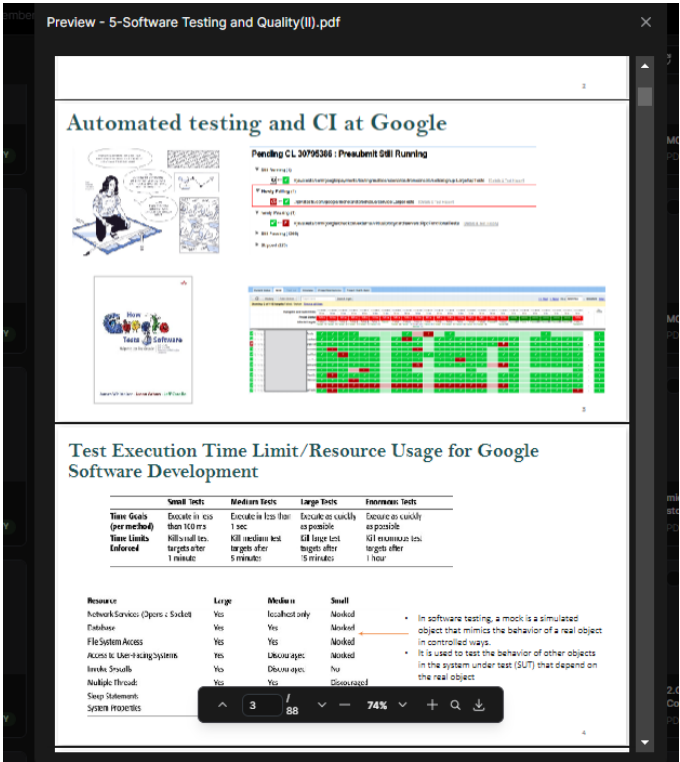
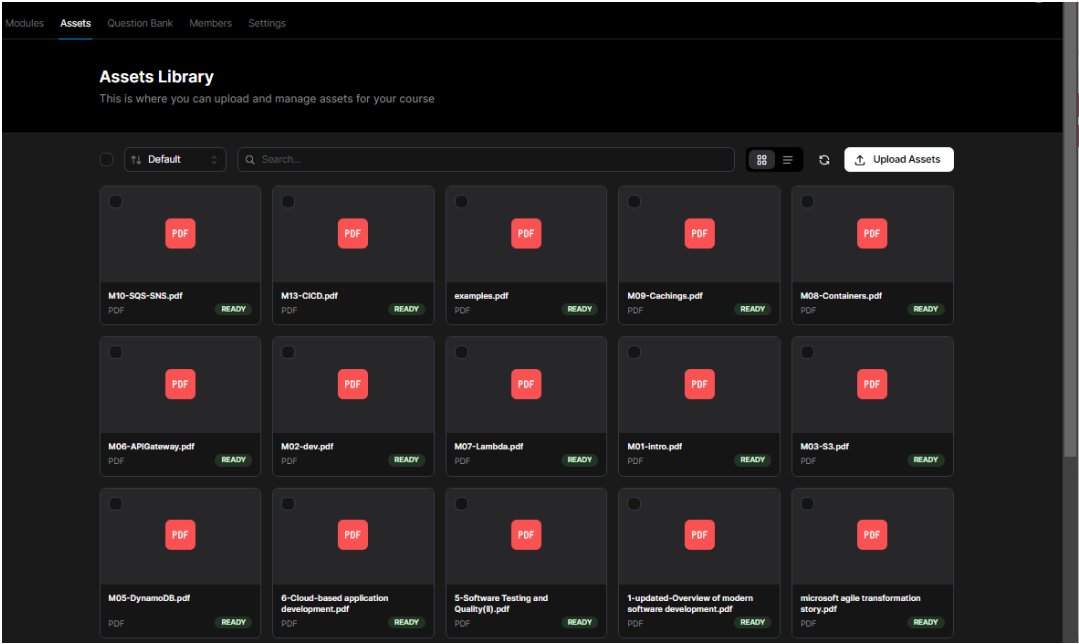
<https://gptutor.comp.polyu.edu.hk>

Platform Features

- Learning Modules



Asset Library



- Student management

ModulesAssetsQuestion BankMembersSettings

Manage Course Members

You can view/add/remove course member here

+ Add Course Member

Add/Drop Member

Search...

☐	Username	Full name	Email	Role	Created at	Edit Role
☐	18		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
☐	19		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
☐	20		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
☐	20		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
☐	20		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
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☐	20		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	
☐	20		@connect.polyu.hk	STUDENT	2024-02-27 12:37 PM	

1 - 10 / 136

Records per page10

1234...14

Students' conversation history

Software Development

SOFTWAREDEVELOPMENT

ModulesLeaderboardConversationsRated Messages

Select date range...

Today

4. Containerized

Filter by member

Search...

Export xlsx

Title	Module	Username	Full Name	Time	View
3. Suppose we want to build a docker image for a Python flask application which	4. Containerized application development	230		2024-03-07 11:15 PM	
According to the "Octoverse" report by GitHub, which of the following are the tw	4. Containerized application development	2201		2024-03-25 08:42 PM	
Agile Software Development and DevOps	4. Containerized application development	2307		2024-02-28 11:02 AM	
Apart from redis, any other open source noSQL is recommend?	4. Containerized application development	2301		2024-03-09 12:39 PM	

Leaderboard

Software Development

SOFTWAREDEVELOPMENT

ModulesLeaderboardConversationsRated Messages

Search...

Export xlsx

Rank	Username	Full name	User Message Count	Conversation count	Last active at	View
1	cswckul	Richard Lui	3164	1616	2024-06-06 01:37 PM	
2			124	7	2024-04-25 03:23 PM	
3			107	7	2024-03-19 02:26 PM	
4			98	23	2024-04-29 07:30 AM	
5			97	52	2024-04-26 01:06 AM	
6			94	41	2024-02-28 09:19 AM	
7			91	11	2024-04-23 04:13 AM	
8			78	29	2024-05-29 04:32 PM	
9			77	22	2024-04-30 05:21 PM	
10			65	31	2024-04-30 09:19 PM	

1 - 10 / 106

Records per page10

1234...11

Exercise generation and practice

Question Bank

This is where you can create and manage questions for your course

WIZARD MODE
Generate Questions >

TEST YOUR KNOWLEDGE
Start a Practice Session >

TEST YOUR KNOWLEDGE
View Flash Cards >

Oldest

Search...

Filter topics

Settings

☆ 🚩 ✎ 🗑

Which of the following options BEST compares one-shot prompting and few-shot prompting?

✗ A. One-shot uses no examples, while few-shot uses one example.

✓ B. One-shot uses one example, while few-shot uses multiple examples.

✗ C. One-shot requires fine-tuning, while few-shot does not.

✗ D. Both require the same number of examples to function.

Answer Explanation

One-shot prompting involves providing one example to guide the model, whereas few-shot prompting involves providing multiple examples.

Topic: One-shot vs Few-shot

☆ 🚩 ✎ 🗑

Which of the following is the BEST example of using Chain-of-Thought prompting?

✓ A. Asking the model to solve a math problem step by step.

✗ B. Requesting a summary of a text without any additional context.

✗ C. Providing a single instruction to generate a poem.

✗ D. Asking the model to write a story without any guidance.

Answer Explanation

Using Chain-of-Thought prompting involves guiding the model to solve a problem step by step, which enhances its reasoning capabilities.

Topic: Practical Application

6

Chat with Course Contents

GPTUTOR / SOFTWAREDEVELOPMENT / Software Testing (II)

Search...

5-Software Testing and...

Guerrilla user testing

- Guerrilla user testing is a low cost method of user testing.
- The term 'guerrilla' refers to its 'out in the wild' style



How to test my web usability? DataCamp

<https://www.youtube.com/watch?v=QY1Qoa5my2I>

What is Usability Testing?

- Tests the ease with which the user interfaces can be used by testing them on users.
- Five components
 - Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
 - Efficiency:** How fast can experienced users accomplish tasks?
 - Memorability:** When users return to the design after a period of not using it, does the user remember enough to use it effectively the next time, or does the user have to start over again learning everything?
 - Errors:** How many errors do users make, how severe are these errors and how easily can they recover from the errors?
 - Satisfaction:** How much does the user like using the system?

Give examples of usability testing

You

Give examples of usability testing

GPTutor

Examples of usability testing techniques include:

- Guerrilla User Testing: A low-cost method conducted in real-world settings to gather user feedback.
- Paper Prototypes: Hand-drawn user interface designs for rapid testing.
- Prototyping Software: Tools like Figma for creating interactive prototypes.
- Conducting Usability Tests: Involves observing users as they interact with interfaces and noting feedback.
- Usability Labs: Setup for electronic observation during usability testing. These examples showcase different approaches to evaluating user experience and interface design.

Here are some ways I can help:

Explain using an example

Explain using an analogy

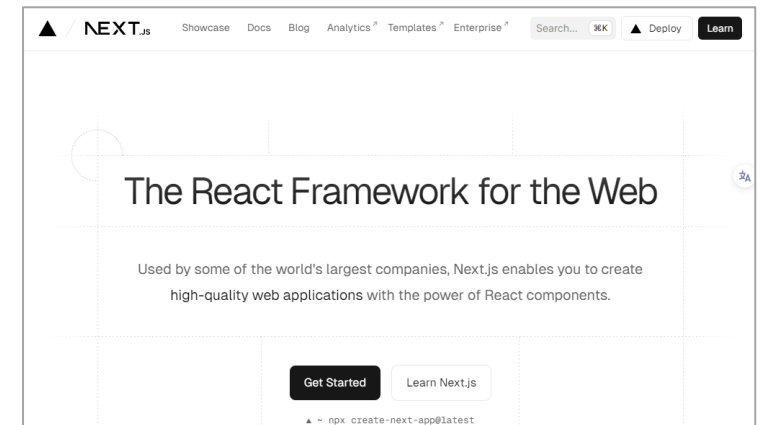
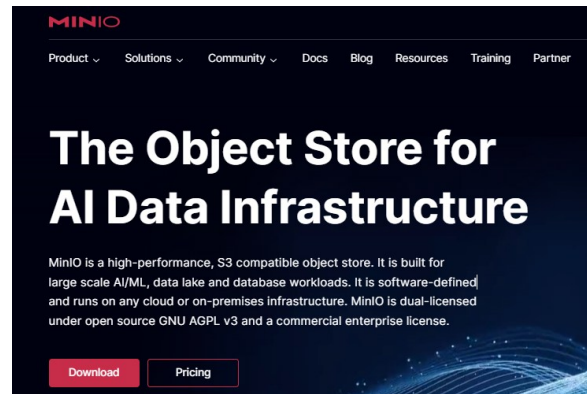
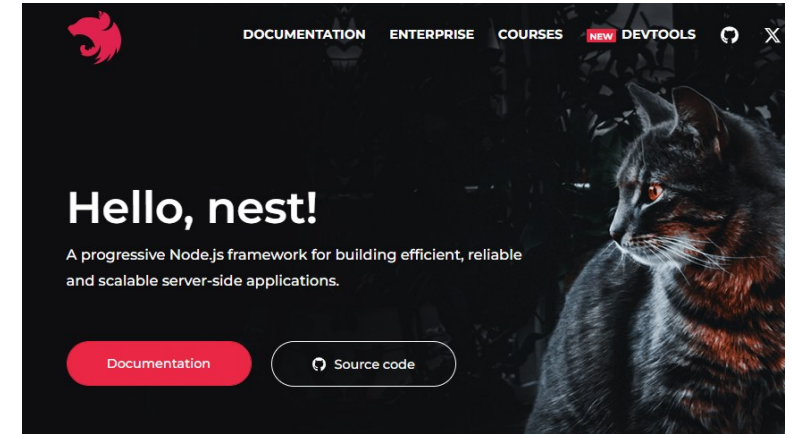
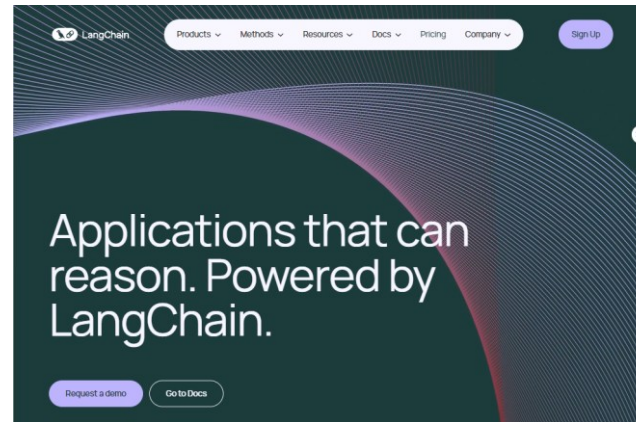
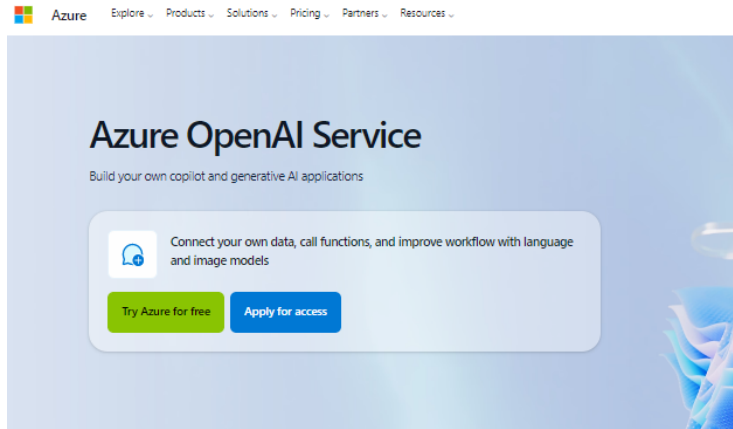
→ What are the components of conducting usability tests?

→ How can incentives improve attendance for usability testing?

→ What is the purpose of paper prototypes in the design process?

Features	Purpose
AI generated response are grounded based on lecture slides	Provide more accurate responses to students' questions with better alignment with the courses' learning outcomes.
Starter questions	Help students initiate conversations and explore the learning materials, even if they are unfamiliar with the topic.
Follow-up questions	Engage students in deeper exploration of the topic.
References to course contents	Allow students to easily access and review the original source lecture notes related to their question, verify the factuality and faithfulness of the AI-generated content.

The Tech Stack of GPTutor



Used as a database and vector store

Knowledge grounded question answering (KGQA)

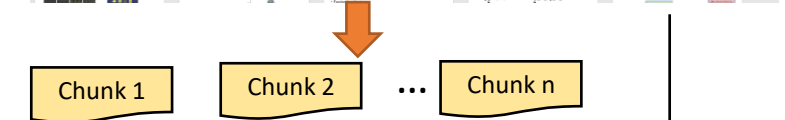
Retrieval-Augmented Generation (RAG): AI technique that improves the quality of a large language model's (LLM) answers by retrieving facts from external knowledge sources and using them as context before generating a response

Embedding: A representation of the semantic meaning of a piece of text

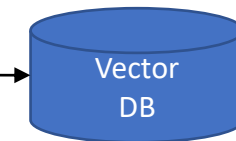
Question:

What is the definition of software crisis

Embedding (question)
[0.692, 0.312, ...]



Create embedding



Retrieve k parts of the slide contents
most relevant to user question



Generate Answer

You
Define software crisis

GPTutor

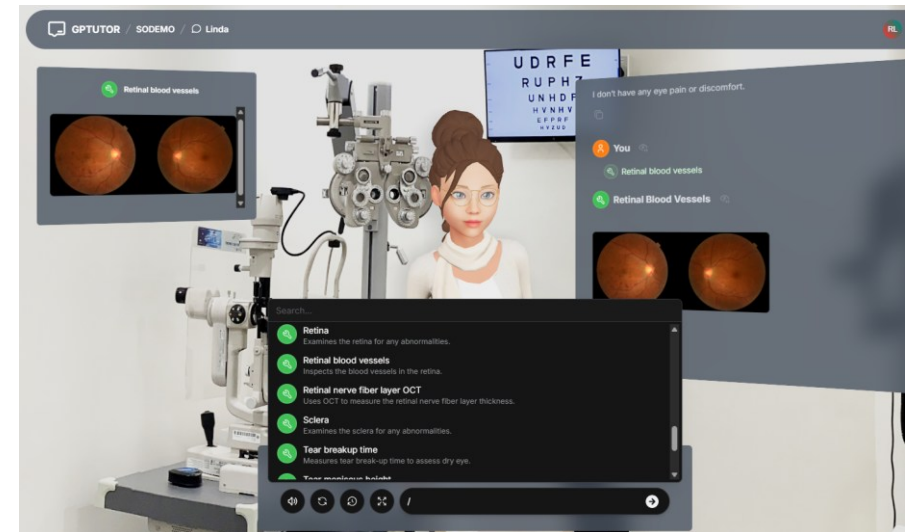
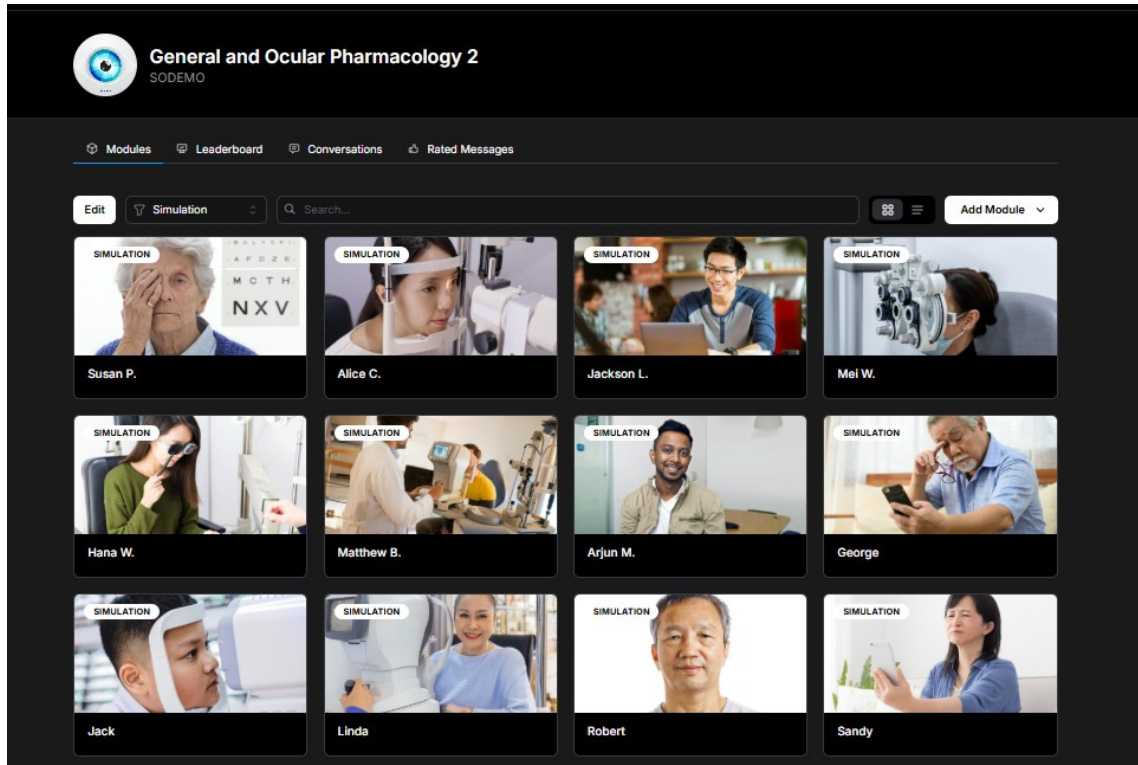
The software crisis refers to the challenges faced in the early days of computing science, such as projects running over-budget, over-time, and delivering inefficient, low-quality software that did not meet requirements. It also manifested in unmanageable projects with difficult-to-maintain code and undelivered software. This led to the coining of the term "software crisis" due to the difficulty of writing useful and efficient computer programs within the required time.

1

- What are the common causes of service disruption?
- What are the key types of non-functional requirements in software development?
- What is the process of requirement elicitation in software development?

Virtual Patient Simulation

- AI-powered virtual patient simulation
 - 3D Avatars
 - Text to speech
- Scenario Builders





Linda

Go to Simulation

Icon

Pick an icon or image to represent this module



Edit

Module Information

Update your module information

Name:

Linda

Description:

Visibility:

AVAILABLE TO STUDENTS

Edit

Background Image

The background image will be used as the backdrop for the simulation.



Edit

Trigger Message

Trigger messages are messages sent by a user to prompt the AI model to respond with a starting message.

 Hello

Answer by

Linda

Edit

Characters

Characters define the various role-plays or personas involved in scenarios. Each character can have unique prompts, background knowledge, and response methods.

Search for a character...



Assistant



Diagnosis Evaluator



Linda

Edit

Actions

Actions are tasks triggered during interactions, with predefined or dynamic responses.

Search for an action...

Amster grid

Uses a grid to detect macular degeneration.

Predefined

Angle depth

Measures the angle between the iris and cornea, important in glaucoma diagnosis.

Predefined

Anterior chamber

Examines the front part of the eye between the cornea and the iris.

Predefined

Blood pressure

Measures blood pressure.

Predefined

Confrontation visual field

Tests visual field using confrontation method.

Predefined

Conjunctiva

Inspects the conjunctiva for signs of infection or other issues.

Predefined

Cornea

Checks the cornea for clarity and abnormalities.

Predefined

Cover test

Identifies misalignment of the eyes (strabismus).

Predefined

Crystalline lens

Checks the lens for clarity and signs of cataracts.

Predefined

Extraocular motility

Assesses the extraocular muscles' ability to move the eyes in different directions.

Predefined

Eyelids

Examines the eyelids for any issues.

Predefined

Habituat spectacles prescription

Determines the current prescription of a patient's regular eyeglasses.

Predefined

Intraocular pressure

Measures intraocular pressure to check for glaucoma.

Predefined

Iris

Inspects the iris for abnormal

Predefined

Edit Action

Cancel Save

Name

Retinal blood vessels

Description

Inspect the blood vessels in the retina.

Answer Mode

Provides a predefined answer

Predefined output

Provide a predefined output that will be shown to the user

